

Clinical: Electrolyte Disorders – Hyponatremia

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Session Objectives

- Distinguish the clinical features and complications of hyponatremia.
- Illustrate the pathophysiology of hyponatremia.
- Judge the causes of hyponatremia.
- Explain the treatment of hyponatremia.
- Explain the syndrome of inappropriate antidiuretic hormone secretion.



Epidemiology

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What percentage of hospitalized patients suffer from hyponatremia?

- A. 6%
- B. 12%
- C. 16%
- D. 22%



Definition

A sodium below which of the
follow levels is categorized as hyponatremia?

- A. 115 mM
- B. 125 mM
- C. 135 mM
- D. 145 mM
- E. 155 mM

At risk populations

- Intensive care unit
- Postoperative
- Psychiatric
- Elderly
- Nursing home patient



Question

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Which of the following recreational drugs has been shown to cause acute hyponatremia?

- A. MDMA (Molly, Ecstasy)
- B. Cocaine
- C. Heroin
- D. Marijuana
- E. None of the above



Question

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What are some potential complications of acute hyponatremia?

- A. seizure activity
- B. brainstem herniation
- C. coma
- D. death
- E. all of the above

Diagnostic Evaluation

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- Look for underlying cause.
- Severe hyponatremia often multifactorial.



History and Physical

- A history of fluid loss
- Detailed medication history
- History consistent with SIADH
- Signs of volume status



Laboratory Tests

- Serum osmolality
- Urine osmolality
- Urine sodium

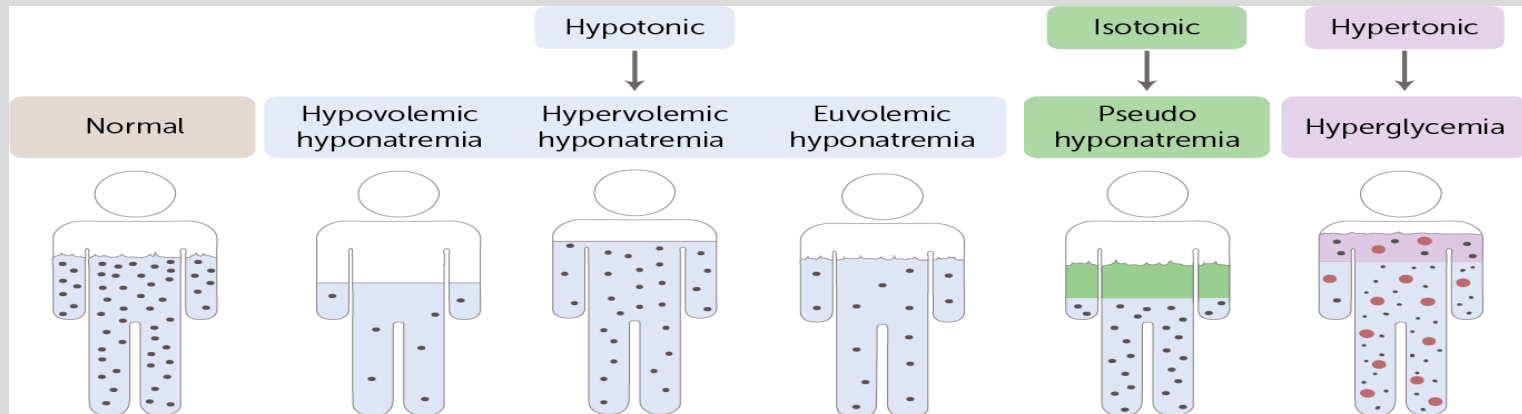
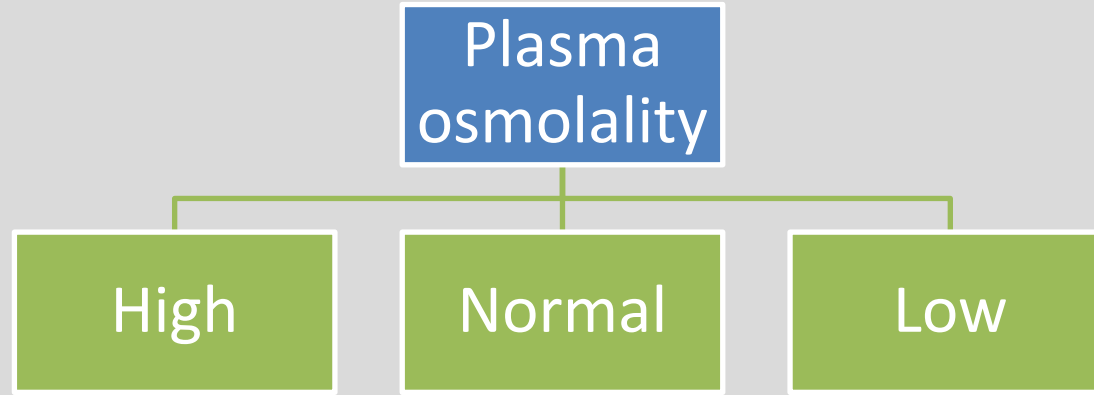
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Hyponatremia Classification

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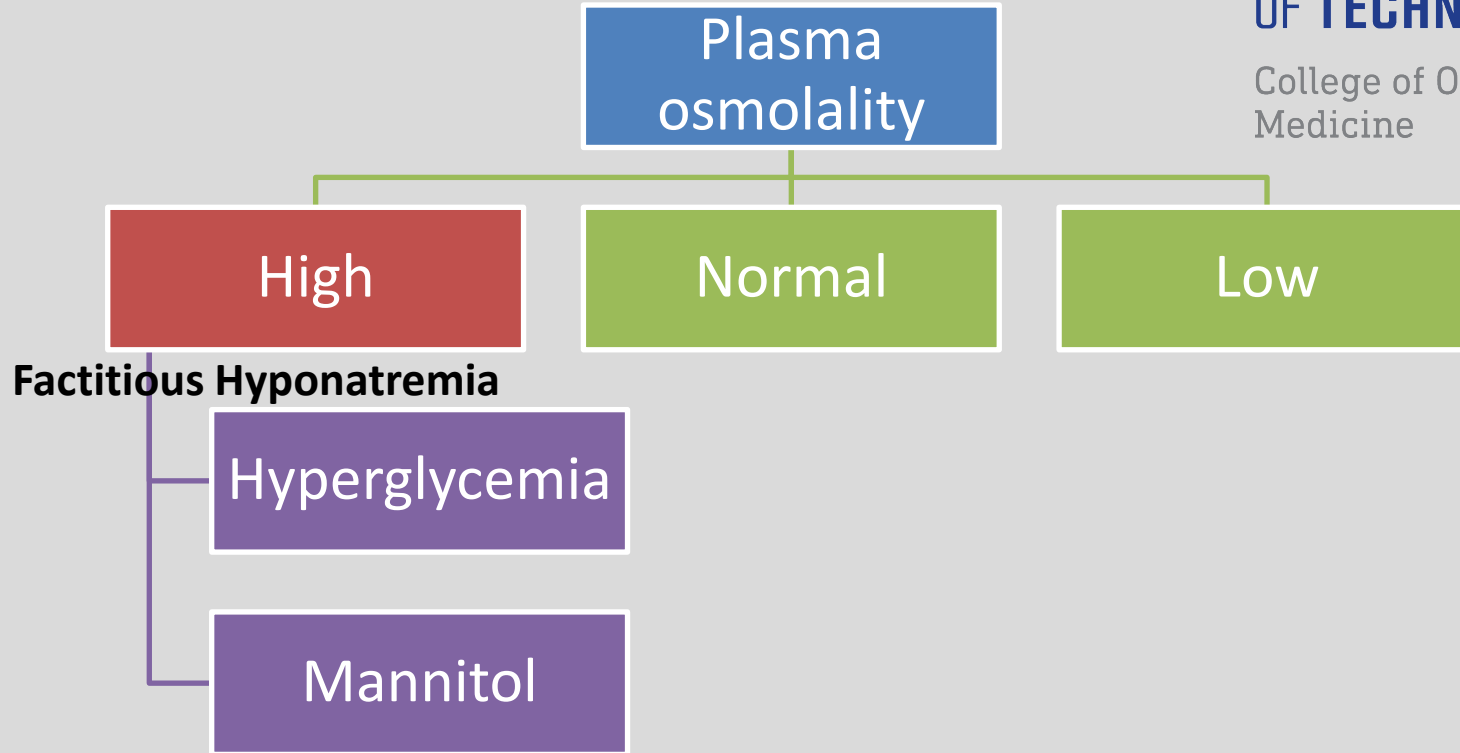




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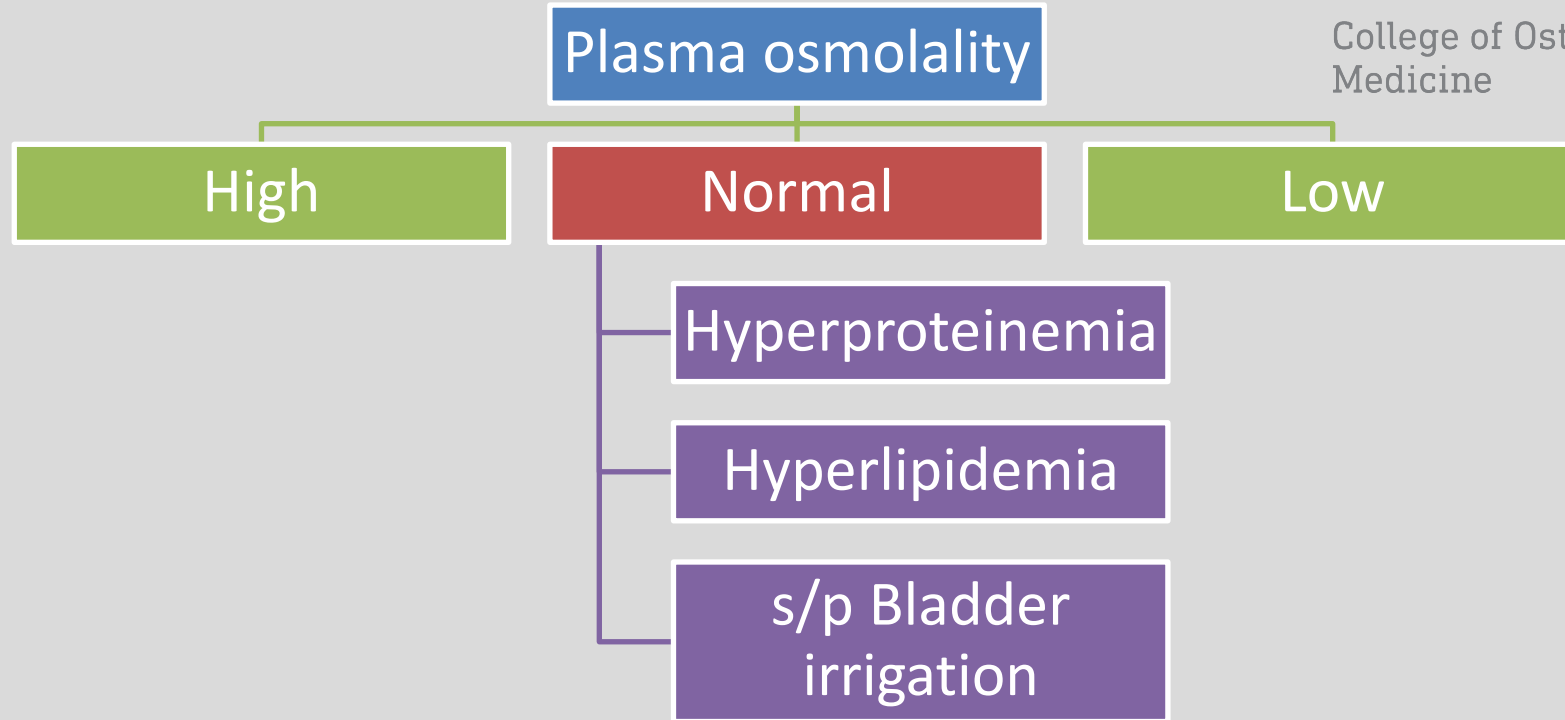




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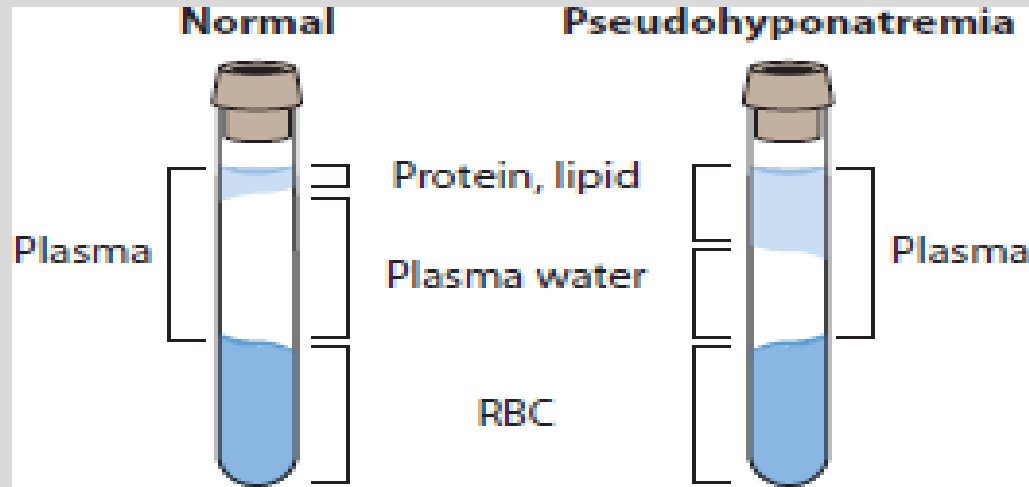




Pseudohyponatremia

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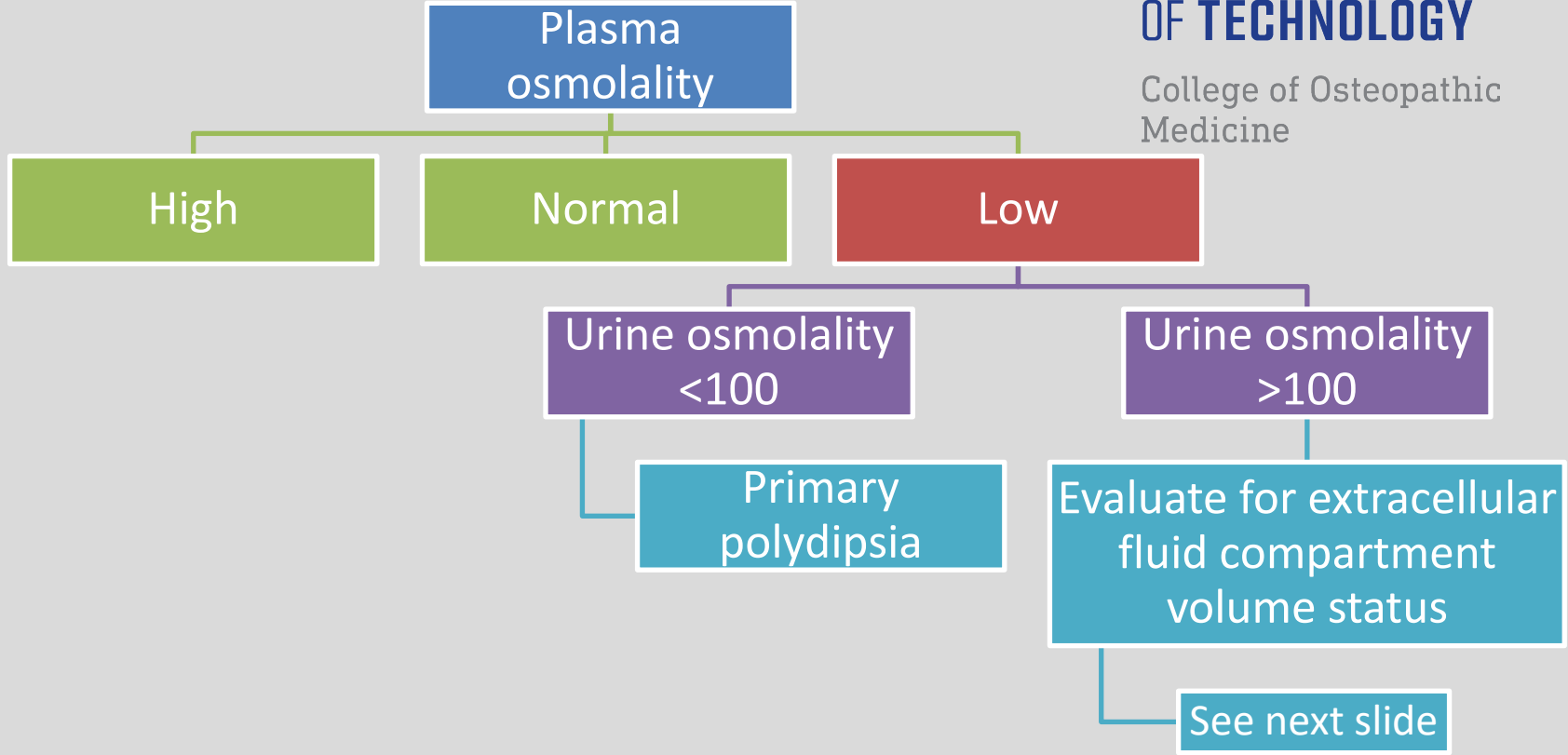
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John P. Kress: *Principles of Critical Care*,
4th Edition: www.accessmedicine.com
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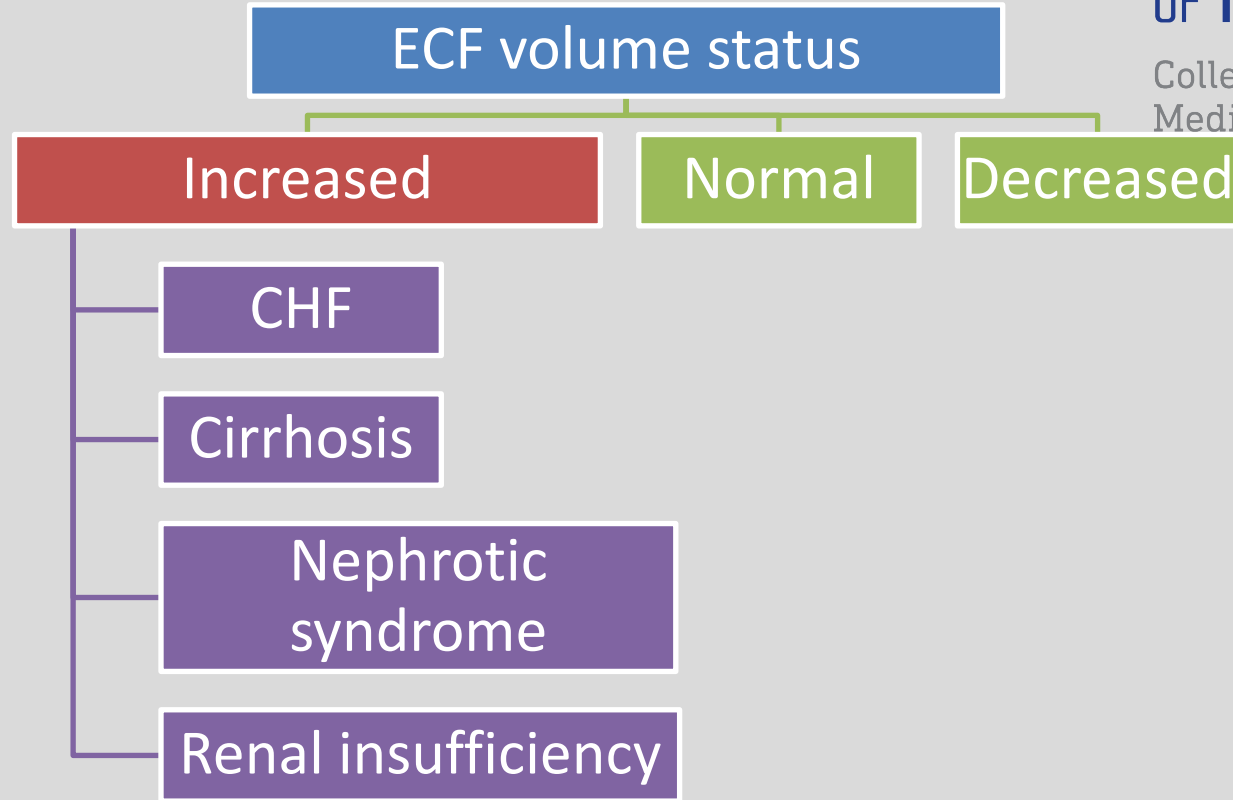




Hypotonic Hyponatremia with: Low Plasma Osmolality

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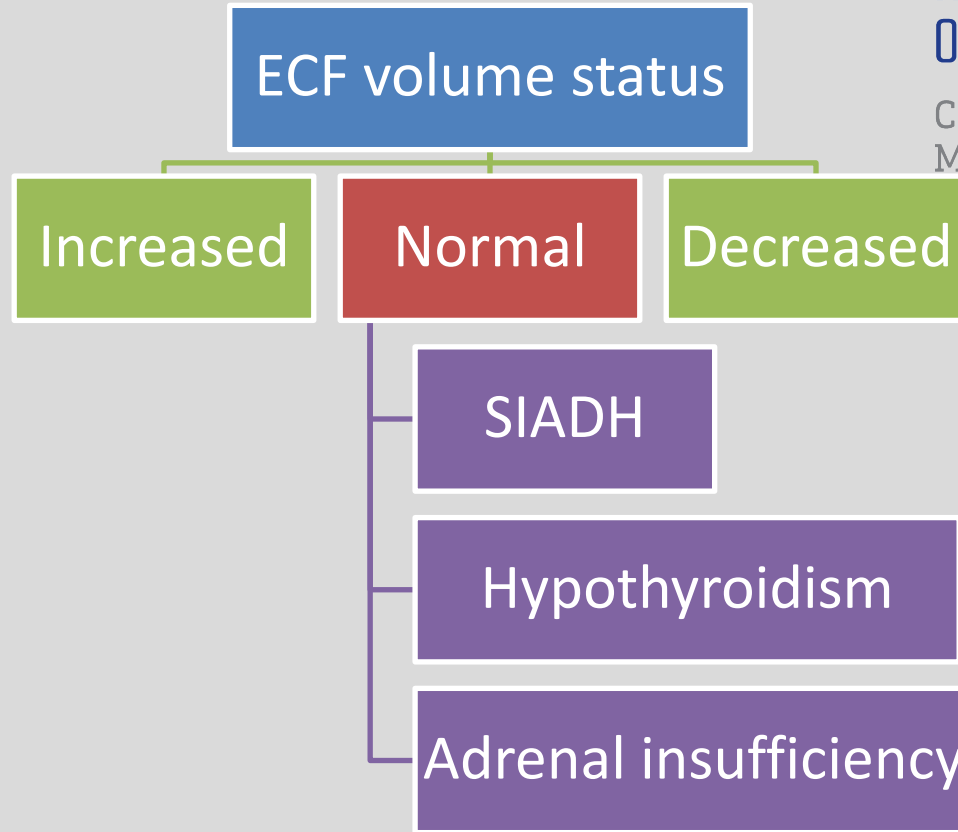




Hypotonic Hyponatremia with: Low Plasma osmolality

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Adrenal Insufficiency

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- Normal to low blood pressure
- Skin hyperpigmentation
- Hyperkalemia

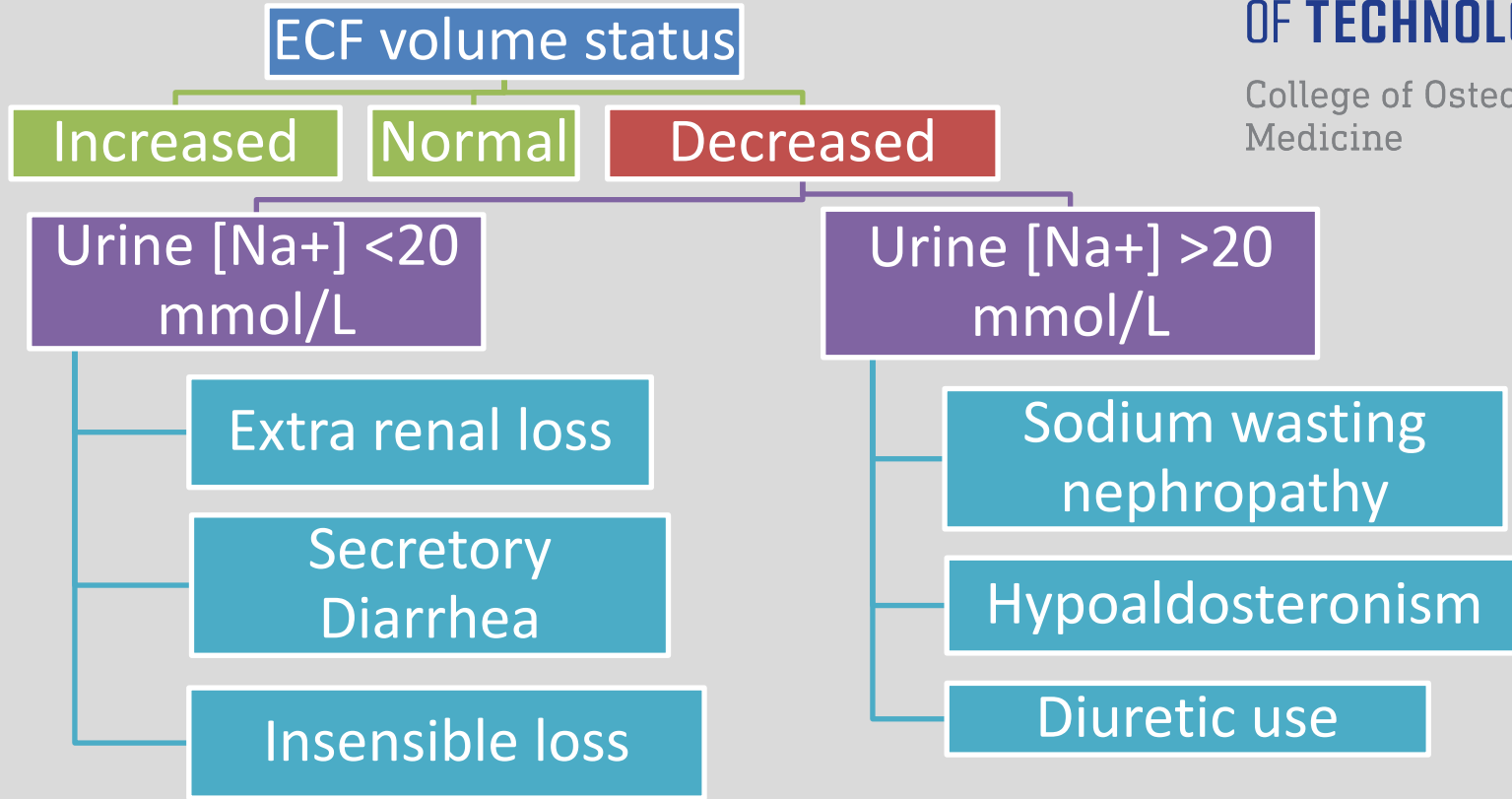




Hypotonic Hyponatremia with: Low Plasma osmolality

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Assessment of volume status

Hypovolemia

- Total body water ↓
- Total body sodium ↓↓

$U_{Na} > 20$

Renal losses
Diuretic excess
Mineral corticoid deficiency
Salt-losing deficiency
Bicarbonaturia with
renal tubal acidosis and
metabolic alkalosis
Ketonuria
Osmotic diuresis
Cerebral salt wasting
syndrome

$U_{Na} < 20$

Extrarenal losses
Vomiting
Diarrhea
Third spacing of fluids
Burns
Pancreatitis
Trauma

Euvolemia (no edema)

- Total body water ↑
- Total body sodium ↔

$U_{Na} > 20$

Glucocorticoid deficiency
Hypothyroidism
Stress
Drugs
Syndrome of inappropriate
antidiuretic hormone
secretion

Hypervolemia

- Total body water ↑↑
- Total body sodium ↑

$U_{Na} > 20$

Acute or chronic
renal failure

$U_{Na} < 20$

Nephrotic syndrome
Cirrhosis
Cardiac failure



Question

A 49-year-old female presents to emergency department with complaints of fatigue, a headache and nausea. Laboratory studies reveal the following: sodium is 129 mEq/L (normal 135-145 mEq/L), serum osmolality is 250 mOsm/kg (normal 275 to 290 mosmol/kg). What test should be next in our workup to differentiate the causes of hyponatremia?

- A. ADH levels
- B. Urine osmolality
- C. Stool culture
- D. Echocardiogram



Question

A 49-year-old female presents to emergency department with complaints of fatigue, a headache and nausea. Physical examination reveals no edema and good skin turgor. Laboratory studies reveal the following: sodium is 129 mEq/L (normal 135-145 mEq/L), serum osmolality is 250 mOsm/kg (normal 275 to 290 mosmol/kg), urine osmolality is 50 mOsm/kg (normal 500-800 mOsm/kg).

Which of the following is the most likely cause of this patient's hyponatremia?

- A. Hyperlipidemia
- B. Primary polydipsia
- C. Hyperglycemia
- D. Cirrhosis
- E. Sodium wasting nephropathy



Question

Which of the following is the most likely pathophysiology behind this patient's hyponatremia?

- A. Increased plasma volume of a substance not contributing to plasma osmolality.
- B. Increased thirst causing water intoxication which overwhelms the body's capacity to excrete water.
- C. Increased plasma osmolality induced by a substance drawing water out of the cells.
- D. A decrease in oncotic pressure resulting in markedly increased vasodilatation and decreased arterial blood volume.
- E. A primary defect in sodium chloride reabsorption in the medullary thick ascending limb of the loop of Henle.

Normal state

- Extracellular fluid osmotic equilibrium with intracellular fluid in brain cells.
- No net movement of water across plasma membrane.

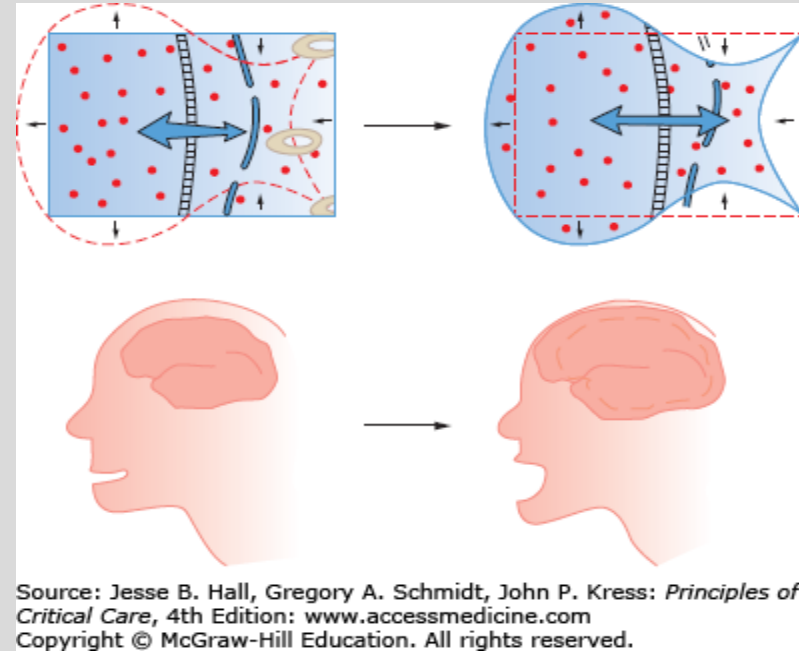
Cerebral Edema

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Acute hyponatremia

- Extracellular fluid hypotonic (lot of water lower solute) compared to intracellular fluid of neurons (less water higher solute).
- Water drawn in by osmosis
- **Cerebral edema via herniation**



Adaptation

- Brain cells pump out osmoles in to the extracellular fluid.
- Osmoles consist first of K^+ and Na^+ salts
- Then organic osmoles
- Creates new osmotic equilibrium

Overly aggressive therapy

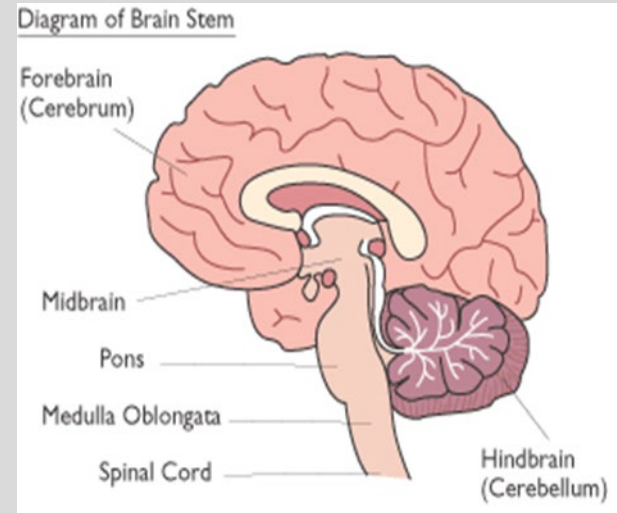
- Hypertonic saline after adaptation occurs.
- Extracellular fluid more concentrated than intracellular fluid.
- Draws water out of brain cells in **osmotic demyelination**.

Cerebral Edema

- Classically affect the pons
- Present 1 or more days after overcorrection
- Quadriparesis, diplopia, loss of consciousness.

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Treatment Hyponatremia

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**Hypertonic and isotonic
hyponatremia treat underlying
cause.**

Hypotonic hyponatremia

- Treat underlying cause
- Determine severity



Treatment Hyponatremia

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- Duration of hyponatremia
 - Acute - first 24 hrs
 - Chronic – more than 48hrs
- Severity of symptoms
 - Or asymptomatic
- Degree of hyponatremia
 - Mild 130-135 meq/L serum sodium
 - Moderate 121-129 meq/L serum sodium
 - Severe 120 meq/L or less.

Setting of treatment

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- Hospitalize – pts with severe hyponatremia and symptomatic moderate hyponatremia
- Outpatient – mild hyponatremia and asymptomatic moderate hyponatremia

Treatment

- Treat based on cause

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Fluid
restriction

Loop
diuretic

NS via IV

Potassium
replacement

Hormone
replacement

Treatment

- Asymptomatic patient
 - Treat with isotonic saline
 - 0.5-1.0 meq/L per hour
 - Raise sodium 4 to 6 meq/L in 24 hour period
 - Do not correct > 10 meq/L in any 24 hour period

Treatment

- Emergent therapy – Aggressive therapy to increase sodium levels if:
 - Severe symptoms
 - Acute symptomatic hyponatremia
 - Self induced water intoxication

Emergent Treatment

- Goal of therapy - increase serum sodium by 4 to 6 meq/L not exceeding 10 meq/L in 24 hours.
- 1-2 mmol/L for 3-4 hours

Outcome based goals

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Hierarchy	Outcomes
Critically important	Patient survival Coma Brain damage/brain oedema Epileptic seizures Osmotic demyelinating syndrome Respiratory arrest Quality of life Cognitive function
Highly important	Bone fractures Falls Length of hospital stay
Moderately important	Serum sodium concentration

Source: Spasovski G, Vanholder R, Allolio B, Annane D, Ball S, Bichet D, Decaux G, Fenske W, Hoorn E, Ichai C, Joannidis M, Soupart A, Zietse R, Haller M, van der Veer S, Van Biesen W, Nagler E; on behalf of the Hyponatraemia Guideline Development Group. Clinical practice guideline on diagnosis and treatment of hyponatraemia. Nephrol Dial Transplant. 2014 Feb 25. [Epub ahead of print] PubMed PMID: 24569496.



Question

An 88-year-old male presents to the emergency department with a complaint of bilateral lower extremity edema. He has a longstanding history of cirrhosis and hyponatremia. The resident notices his sodium level is 119 mEq/L (normal 135 and 145 mEq/L) and administers copious hypertonic saline. Two days later the patient is comatose and eventually dies. On autopsy which of the following brain structures is most likely to be affected?

- A. Pons
- B. Cerebellum
- C. Medulla
- D. Pre-frontal cortex
- E. Parietal lobe

Case scenario

- 78-year-old man was admitted with pneumonia and hyponatremia.
- Plasma Na⁺ concentration was initially 129 meq/L, decreasing within 3 days to 118–120 meq/L despite fluid restriction to 1 L/d. A chest CT revealed a right 2.8 × 1.6 cm infrahilar mass and postobstructive pneumonia.
- The patient was an active smoker. Past medical history was notable for laryngeal carcinoma treated 15 years earlier with radiation therapy, renal cell carcinoma, peripheral vascular disease, and hypothyroidism.

Case scenario

- On review of systems, he denied headache, nausea, and vomiting. He had chronic hip pain that was managed with acetaminophen with codeine.
- Other medications included cilostazol, amoxicillin/clavulanate, digoxin, diltiazem, and thyroxine.
- He was euvolemic on examination, with no lymphadenopathy and a normal chest examination.

Case scenario

Labs

- Albumin – 3.4 g/dL (3.4 to 5.4 g/dL)
- Na⁺ - 125 mEq/L (135 and 145 mEq/L)
- Cortisol – 25 ug/dL (10 to 20 ug/dl)
- BUN – 8 mg/dL (7 to 20 mg/dL)
- Cr – 1 mg/dL (0.84 to 1.21 mg/dL)
- Glucose – 93 mg/dL (70-99 mg/dL)
- Plasma osmolality – 250- mosmol/kg (275 to 290 mosmol/kg)
- Urine osm – 890 mOsm/kg (500-800 mOsm/kg).

Diagnosis?

SIADH

- Impaired water excretion
- Inability to suppress ADH
- ADH causes concentrated urine increase total body water.
- Low plasma sodium concentration

SIADH Presentation

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Fatigue Headache Malaise

Confusion
Altered mental state

Nausea
Vomiting
Abdominal pain

⚠ Hypoosmolality ⚠ Hyponatremia

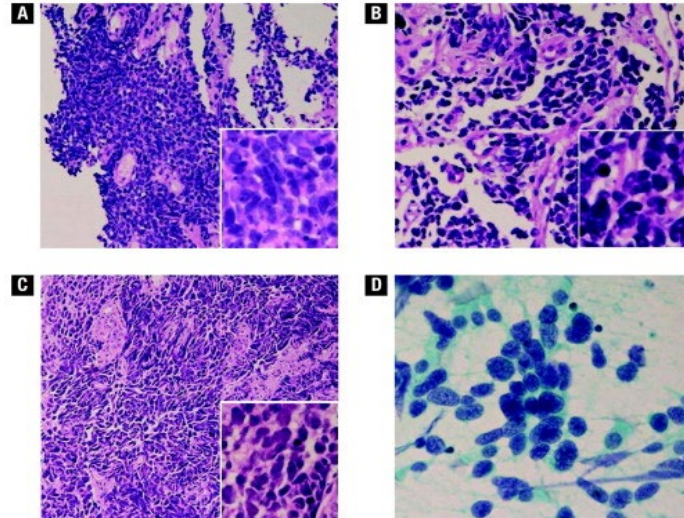


Common Etiologies

- Malignancies
- Drugs
- Surgery
- Idiopathic
- Hereditary

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Kurahara Y, Kawaguchi T, Tachibana K, et al. Small-cell lung cancer in never-smokers: a case series with information on family history of cancer and environmental tobacco smoke. *Clin Lung Cancer*. 2012;13(1):75-79. doi:10.1016/j.clcc.2011.04.001

Treatment

- Fluid restriction – mainstay maintenance therapy (800-1000mL/day)
- Combined salt tablets and furosemide (loop diuretic)
- Vasopressin receptor antagonists
- Correct less than 10meq/L in 24 hours

Key Points

- Key points in management of the patient with hyponatremia
 - Diagnose the cause
 - Clinical history / findings / lab test
 - Determine a treatment plan
 - How to treat
 - How fast to treat
 - What to monitor
 - Assess need for follow up
 - When do you expect improvement
 - When do you need to next evaluate the patient
 - Anticipate complications from treatment



Question

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A 55-year-old female presents to ED with complaints of fatigue, a headache and nausea. PE reveals no edema and good skin turgor. Lab studies reveal sodium is low, serum and urine osmolality are normal. What is the diagnosis?

- A. Hyperlipidemia
- B. Hyperglycemia
- C. Cirrhosis
- D. Hypothyroidism

Summary Slide

- Distinguish the clinical features and complications of hyponatremia.
- Illustrate the pathophysiology of hyponatremia.
- Judge the causes of hyponatremia.
- Explain the treatment of hyponatremia.
- Explain the syndrome of inappropriate antidiuretic hormone secretion.
- Core References:
 - Mount DB. Fluid and Electrolyte Disturbances. In: Longo D, Fauci A, Kasper D, Hauser S, Jameson J, Loscalzo J, Holland S, Langford C. eds. Harrison's Principles of Internal Medicine, 22nd Edition. McGraw Hill; 2026
 - Sterns. R. (2025). Causes of hyponatremia in adults. *UpToDate*.
 - Sterns. R. (2024). Overview of the treatment of hyponatremia in adults. *UpToDate*.

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